

Parthiban S

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PROFESSIONAL SUMMARY

Electrical and Electronics Engineering student with hands-on experience in embedded systems, MATLAB/Simulink modeling, and renewable energy systems (solar PV, MPPT). Strong foundation in power system protection, switchgear, and control systems, with practical exposure to circuit diagnostics and hardware troubleshooting. Seeking to apply skills in electrical product engineering, panel commissioning, and O&M to deliver reliable, standards-compliant engineering solutions.

EDUCATION

B.E. Electrical and Electronics Engineering (Pursuing) — 2023–2027

V.S.B. College of Engineering, Coimbatore (Anna University) — CGPA: 7.94/10

Higher Secondary — 2022–2023

Palanigounder Hr. Sec. School — 65%

SSLC — 2020–2021

RKR Gnanodhaya Matric. Hr. Sec. School

PROJECTS

Dual-Axis Solar Tracking System

- Designed and implemented a dual-axis solar tracking system using Arduino Nano, LDR sensors, and servo motors; automated real-time panel alignment with the sun's position, improving energy efficiency by ~35% over fixed-panel installations.
- Modeled system behavior and validated control logic using MATLAB/Simulink prior to hardware deployment.
- Integrated IoT (NodeMCU/ESP8266) for remote monitoring of panel orientation and output data.
- Tech Stack: Arduino Nano, LDR sensors, Servo motors, Embedded C, IoT, MATLAB/Simulink
- GitHub: github.com/parthi005/solar-tracking-system

LED Board Display using Arduino

- Developed a programmable LED display board with scrolling text messages via Arduino-based embedded C control logic.
- Designed and documented circuit interfacing, wiring schematics, and hardware layout.

WORK EXPERIENCE

Study Shinee Software Solution — Embedded Systems Intern | Dec 2025

- Gained hands-on experience with I2C/UART communication protocols and firmware development for microcontroller-based systems.
- Applied embedded systems fundamentals to real hardware-software integration tasks under mentorship.

VVDN Technologies, Pollachi — Hardware Intern (Laptop Assembling & Circuit Debugging) | Aug 2024

- Performed PCB-level diagnostics and circuit troubleshooting across laptop hardware assembly lines.
- Identified and resolved hardware faults, contributing to a reduction in component failure rate during quality checks.

CERTIFICATIONS

- NPTEL (Elite) — Internet of Things, IIT Kharagpur — 2025
- NPTEL (Elite) — Industrial Internet of Things 4.0, IIT Kharagpur — 2025

ACHIEVEMENTS

- Presented "Renewable Resources using Micro-Grids" at Smart India Hackathon — Intern College, 2025.
 - Co-authored and published "THERMO-LYTIC: Multi-Source Hybrid for Green Hydrogen Production" — IEEE Conference, 2025.
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KEY SKILLS

Programming Languages: Java (Methods, Arrays, String, OOPS), Embedded C/C++

Simulation & Design Tools: MATLAB, Simulink, Arduino IDE, VS Code, Proteus (basic)

Protocols & Embedded: I2C, UART, Microcontroller/Microprocessor interfacing, Firmware development

Electrical Concepts: Power System Analysis, Transmission & Distribution, Power System Protection & Switchgear, Power Electronics, MPPT, Solar PV Systems, Micro-Grid, IoT

Soft Skills: Communication, Teamwork, Time Management, Leadership

AREAS OF INTEREST

- Solar Energy Systems (PV Design & Installation, MPPT)
- Renewable & Non-Renewable Energy Integration
- Transmission & Distribution
- Power System Analysis
- Protection & Switchgear

LANGUAGES KNOWN

English, Tamil